

CMPE – 211

Preliminary Work (Pre-Lab Activity)

Laboratory Experiment #3

Textbook Material: Chapter 6 «Methods»

• • • TASK 1

Write, Compile and Execute a **Java method** that computes the sum of the digits in an integer. Use the following method header:

```
public static int sumDigits(long n)
```

For example, `sumDigits(234)` returns **9** ($2 + 3 + 4$). (Hint: Use the `%` operator to extract digits, and the `/` operator to remove the extracted digit. For instance, to extract 4 from 234, use `234 % 10` ($= 4$). To remove 4 from 234, use `234 / 10` ($= 23$). Use a loop to repeatedly extract and remove the digit until all the digits are extracted. Write a test program that prompts the user to enter an integer and displays the sum of all its digits.

• • • TASK 2

Write, Compile and Execute the **Java methods** with the following headers to check if a number is a palindrome:

```
// Return the reversal of an integer
```

```
public static int reverse(int number)
```

```
// Return true if number is a palindrome
```

```
public static boolean isPalindrome(int number)
```

For example, `reverse(3456)` displays **6543**. Use the `reverse` method to implement `isPalindrome`. A number is a palindrome if its reversal is the same as itself. Write a test program that prompts the user to enter an integer and reports whether the integer is a palindrome.

• • • TASK 3

In number theory, a **happy number** is a number which eventually reaches 1 when replaced by the sum of the square of each digit. For instance, 13 is a happy number because

$$1^2 + 3^2 = 10 \text{ and } 1^2 + 0^2 = 1.$$

On the other hand, 4 is not a happy number because the sequence starting with

$$4^2 = 16 \text{ and } 1^2 + 6^2 = 37$$

and eventually reaches to 4 that is the number that started the sequence, and so the process continues in an infinite cycle without ever reaching 1.

Write, Compile and Execute a **Java method** that computes whether a given number is happy or sad.

• • • Sources

- Introduction to Java Programming and Data Structures, Comprehensive Version, 11th Edition, Y. D. Liang, Pearson, 2019.